

Stockd.

Our project, “**Stockd.**”, is a high-level food management system that detects and tracks what a user puts in and takes out of their refrigerator. The scanner detects what the item is and updates the users on its expiration date if applicable. For items like fresh produce, it will give the user an estimated date to consume/use the item by; for items like leftovers, it will allow users to estimate an expiration date. Stockd can also be used without the camera attachment if someone wants to manually log the items in their kitchen using the connected app and their phone camera, but the scanner for the fridge makes tracking much more user friendly and convenient.

The most comparable developed program is Samsung’s Family Hub which keeps track of items in their smart fridge. It allows users to look inside their fridge from anywhere using an app on their phone to track stock and expiration dates. However, their application requires users to buy their fridge while our project is an addition to an already existing fridge and potentially could be used in a pantry as well. Our app would have a variety of features to help users manage their fridge such as: expiration date adder/tracker, custom meal recommendations based on what's in the user’s food database, lists of current inventories, grocery shopping suggestions based on usage, etc. This project will primarily use Python and OpenCV for computer vision and Raspberry Pi processing and React native for the application. We plan to also use Supabase for database storage and synchronization, and potentially WebSockets or another real-time communication method. For hardware, we plan to use Raspberry Pi Camera and Raspberry Pi 5.

Our product is designed to be used for any household, however, it is especially helpful for college students or other people who are busy and have difficulty tracking their groceries and leftovers. Food is often forgotten about in pantries and fridges until it expires which is a large contributor to food waste and additional grocery expenses. Stockd will make users' everyday lives more convenient by helping them decide what food to use first, what needs to be re-bought, and overall reduce their food waste.

Our team’s choice for this project is appropriate for this class because not only will it be an opportunity for us to showcase what we have learned, it will also allow us to work with some hardware. We will be able to work on building a website/app, working with an API, coding in Python, and connecting the different parts together to create a cohesive product; which are all things we have learned how to do. We will be challenging ourselves to work with new software and hardware, as we haven’t worked with Raspberry Pi or OpenCV. The reason we wanted to work on this project is because all of us believe that this would be a product that will help us and many other people with reducing food waste and keeping track of perishable items.